



Advanced Internet of Things Technology

Week 7: Open Research Directions and Open-Source Platforms in IoT

TS. Huỳnh Văn Đăng

Department of Computer Networks

Faculty of Computer Networks and Communications

University of Information Technology, VNUHCM

Email: danghv@uit.edu.vn

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Motivation: Why IoT Is Still an Open Research Field

- IoT is evolving from simple sensor networks into a global cyber-physical infrastructure
- Integration of AI, edge computing, and advanced wireless networks is reshaping IoT systems
- Massive data generation requires distributed intelligence instead of cloud-only processing

Learning Objectives

- Identify major emerging research directions in IoT systems
- Understand how edge intelligence and 6G reshape IoT architecture
- Compare open-source hardware and software ecosystems for IoT
- Understand practical development stacks used in modern IoT systems

Part I: Open Research Directions in IoT

Evolution of IoT Toward Intelligent Systems

From IoT to Internet of Everything

- Traditional IoT architecture focuses on sensing, connectivity, and cloud analytics
- Modern IoT integrates edge intelligence and distributed computing
- Autonomous decision making is increasingly embedded in devices and edge nodes

Convergence with Future Networks

- Next-generation wireless systems enable ultra-massive device connectivity
- Real-time immersive applications require extremely low latency
- Integrated sensing and communication supports environment awareness
- Distributed AI across the network supports intelligent decision making

Topic 1: AI/ML for IoT

AI-Native IoT and Edge Intelligence

- Cloud-only processing cannot satisfy latency requirements of many IoT applications
- Edge intelligence enables real-time analytics close to data sources
- Local processing improves privacy and reduces communication overhead

TinyML

- Machine learning inference on microcontrollers and embedded devices
- Neural networks optimized for extremely limited memory and power
- Efficient inference through quantization and model compression

Research Challenges

- Hardware-aware machine learning models
- Adaptive learning on edge devices
- Managing large fleets of intelligent devices

3 generative AIoT applications

1 Code generation for IoT

Large language models can be used to create, complete or combine IoT code, sourced from code snippets or natural language descriptions.

Example



GitHub Copilot



 GitHub Copilot

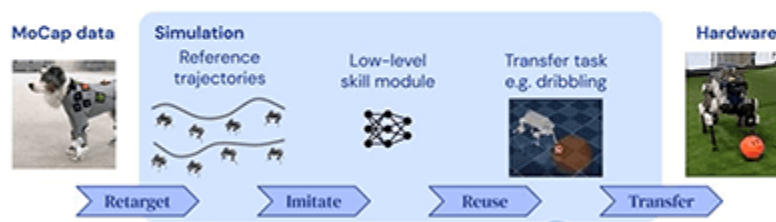
2 Robot control

Generative AI can be used to generate controls and commands of IoT connected robots by capturing motion data from animals or humans.

Example



Deepmind Robot Control Framework



 DeepMind

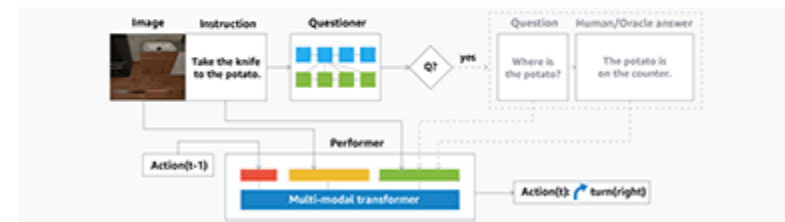
3 Social IoT devices

Generative AI can be used to change the interaction with devices. Devices, for example, could provide the user with answers to complex question.

Example



Amazon DialFRED



 | science



Motivation

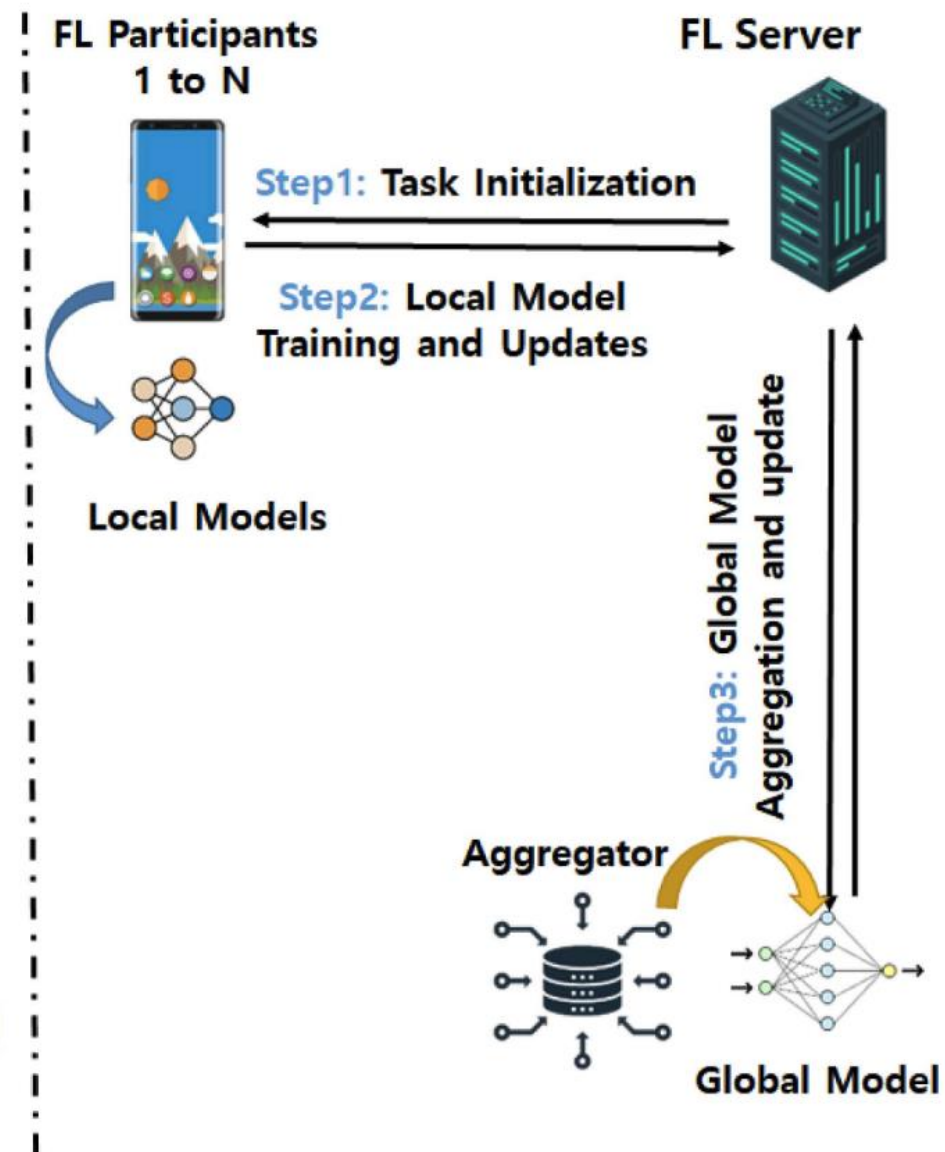
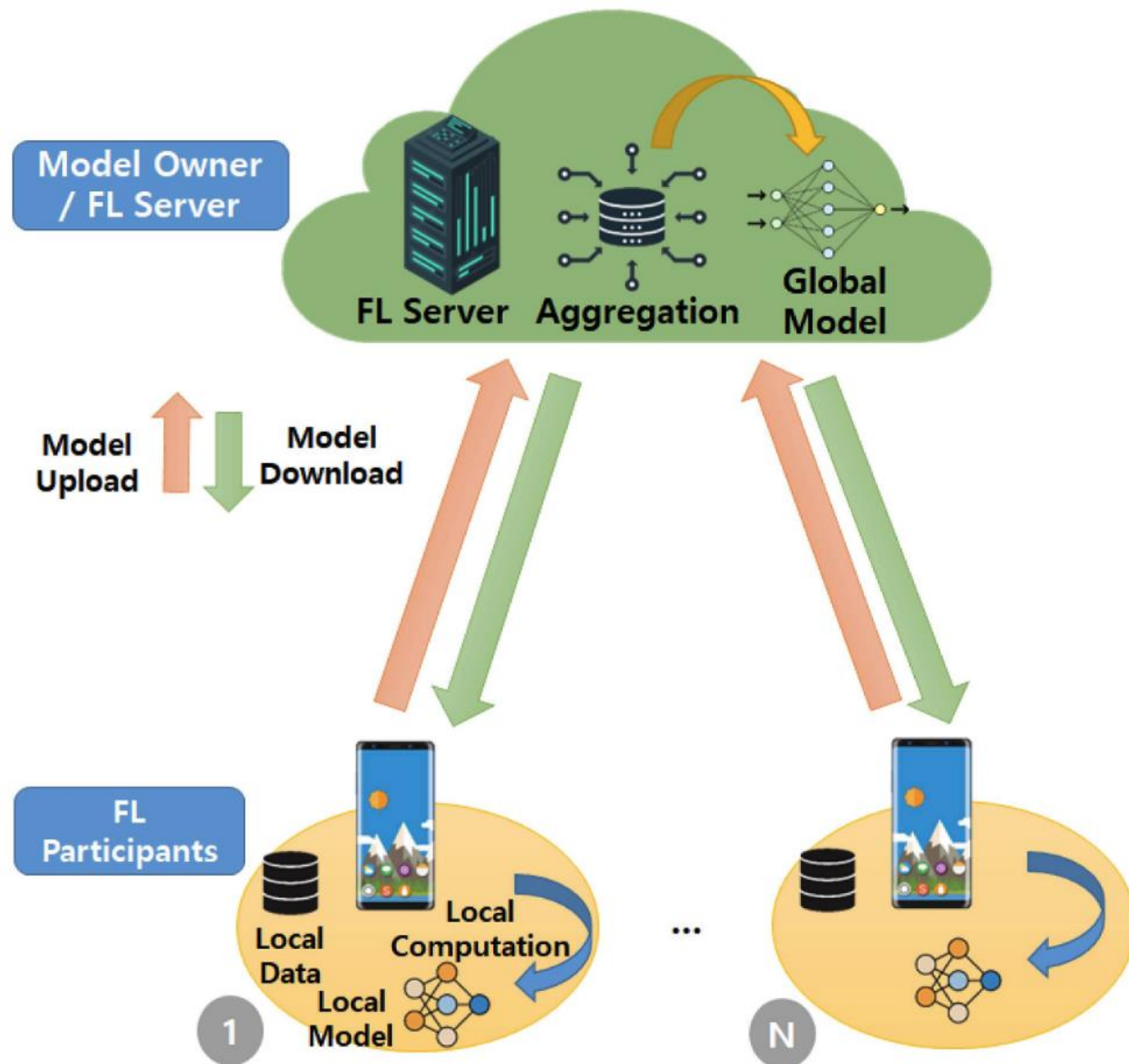
- IoT data is distributed across devices and organizations
- Centralized training creates privacy and communication challenges

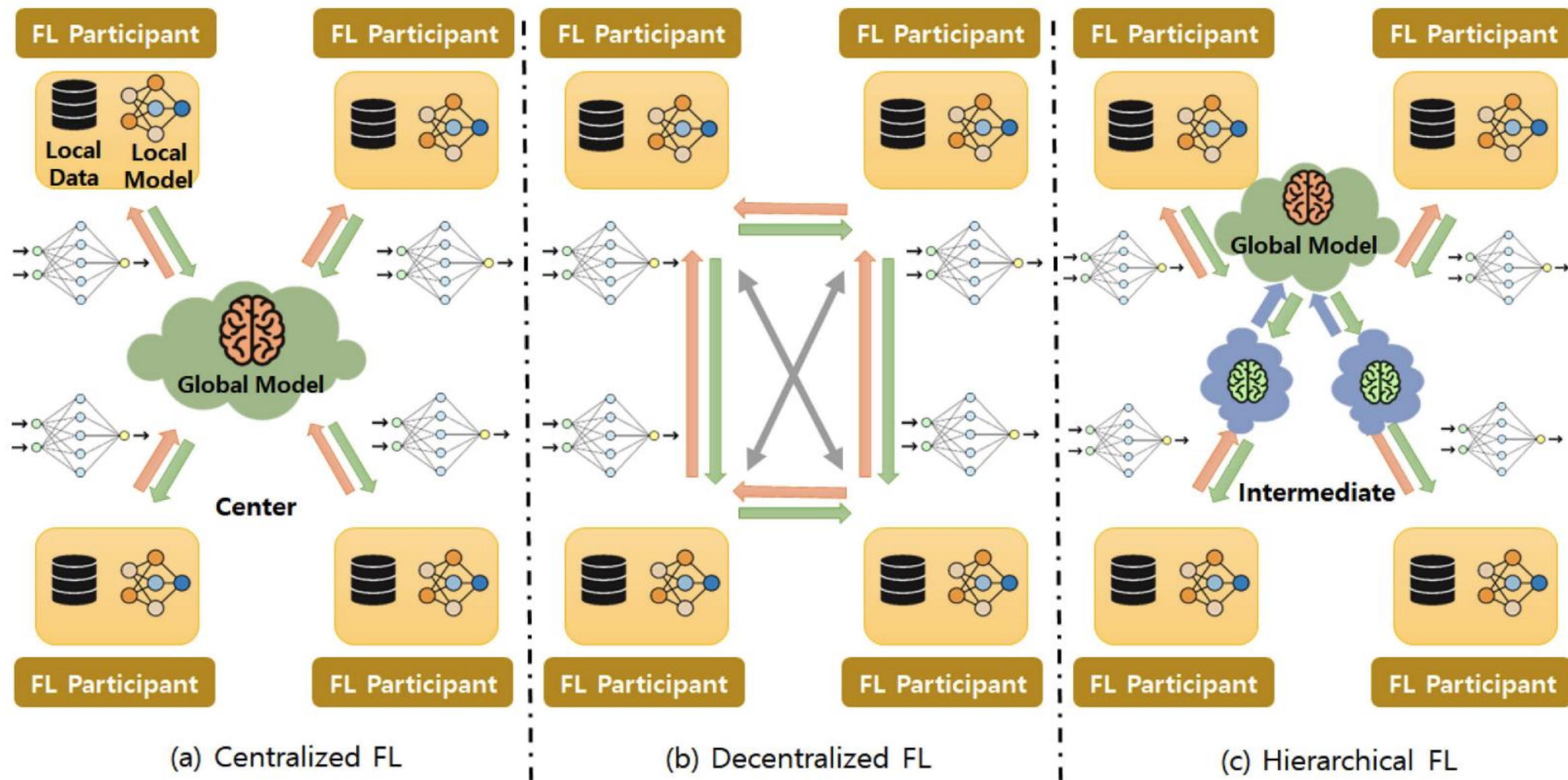
Federated Learning Workflow

- Devices train models locally
- Only model updates are transmitted to aggregation servers
- Global models are built without sharing raw data

Key Challenges

- Non-IID data distributions across devices
- Communication overhead during training
- Device heterogeneity and unreliable connectivity





Topic 3: Digital Twins for IoT Systems



Digital Twin Concept

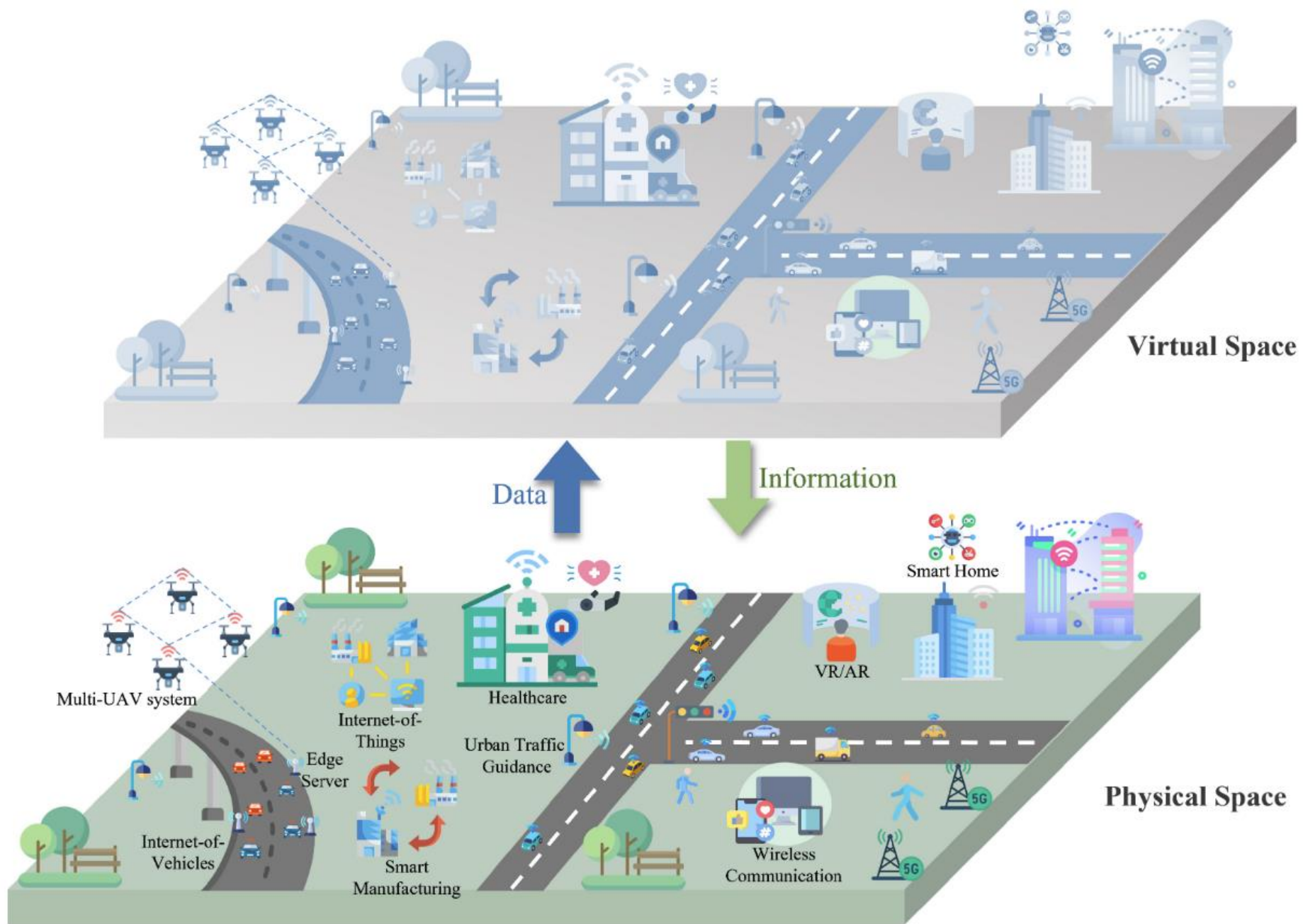
- Virtual representation of physical systems
- Continuous synchronization with real-world operations
- Enables monitoring, prediction, and system optimization

Applications

- Smart cities and infrastructure monitoring
- Industrial automation and predictive maintenance
- Network performance optimization

Research Challenges

- Accurate modeling of complex systems
- Real-time synchronization between physical and digital environments
- Scalability to very large IoT deployments



Topic 4: IoT Security and Zero Trust Architecture



Security Challenges

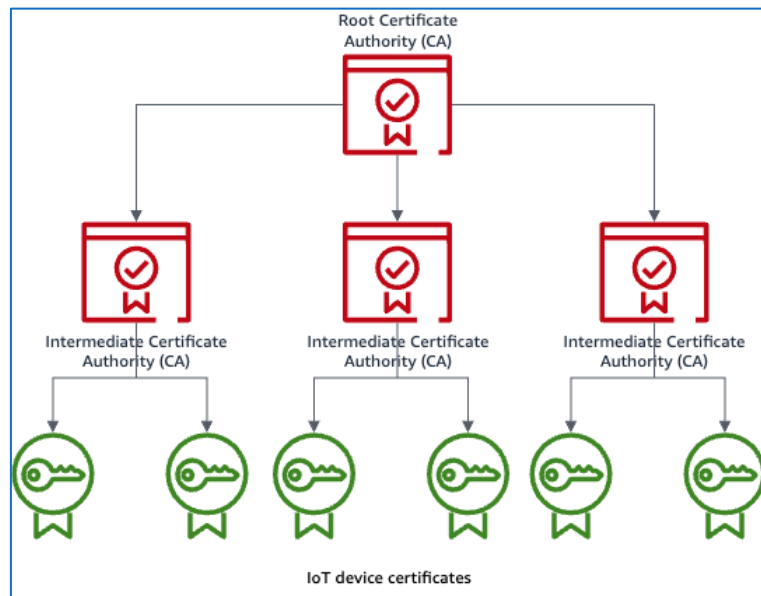
- Weak authentication mechanisms
- Outdated firmware and poor lifecycle management
- Large attack surfaces due to massive device deployments

Zero Trust Principles

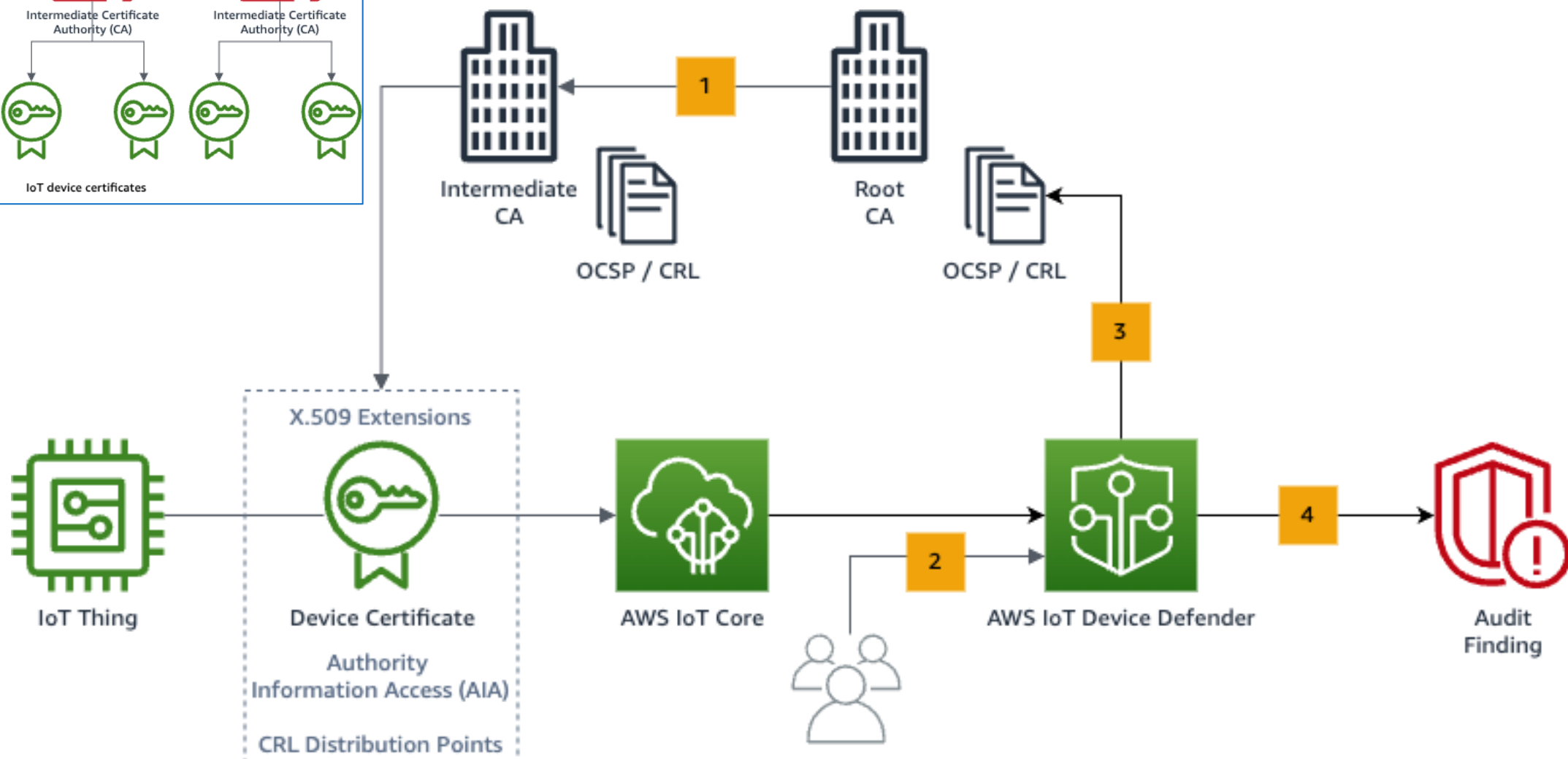
- Continuous authentication of devices and users
- Least privilege access control
- Micro-segmentation of networks

Machine Identity

- Unique cryptographic identity embedded into devices
- Automated certificate management
- Secure machine-to-machine communication



<https://aws.amazon.com/blogs/iot/identifying-iot-device-certificates-with-a-revoked-intermediate-ca-using-aws-iot-device-defender/>



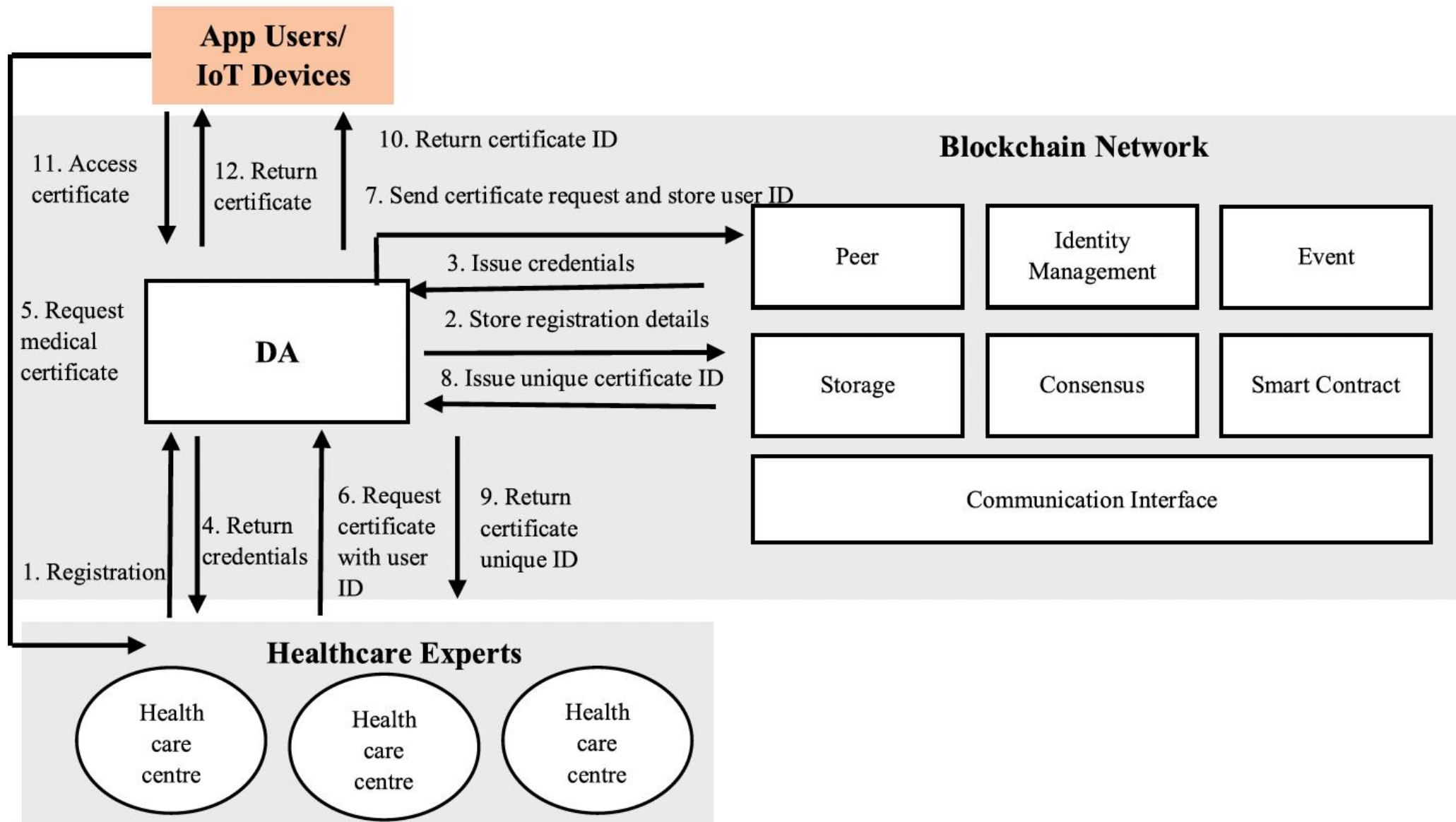


Blockchain Applications

- Decentralized trust among IoT devices
- Tamper-proof transaction records
- Smart contracts for automated interactions

Privacy-Preserving Techniques

- Federated learning
- Differential privacy
- Trusted execution environments
- Homomorphic encryption
- Zero-knowledge proofs





Motivation

- Large IoT deployments significantly increase energy consumption
- Sustainability becomes a major design requirement

Energy-Harvesting IoT

- Solar energy harvesting
- RF energy harvesting
- Mechanical vibration energy harvesting

Carbon-Aware Networking

- Event-driven sensing reduces unnecessary transmissions
- Energy-aware routing decisions
- Communication protocols designed for sustainability

Sustainable Smart City





THz Communications

- Extremely high bandwidth communication links
- Enabling ultra-high data rate applications

Reconfigurable Intelligent Surfaces

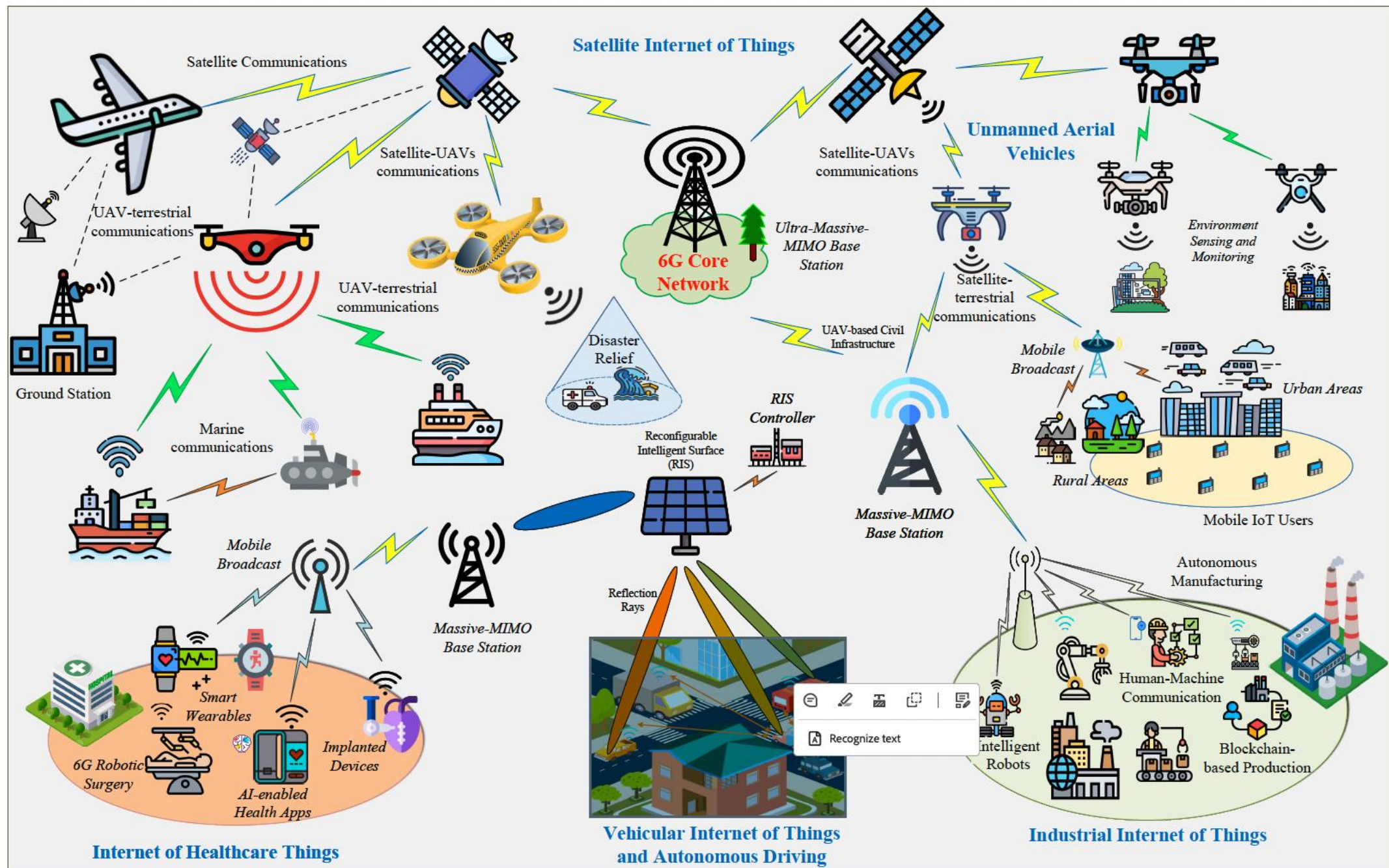
- Programmable wireless environments
- Improved signal propagation and coverage

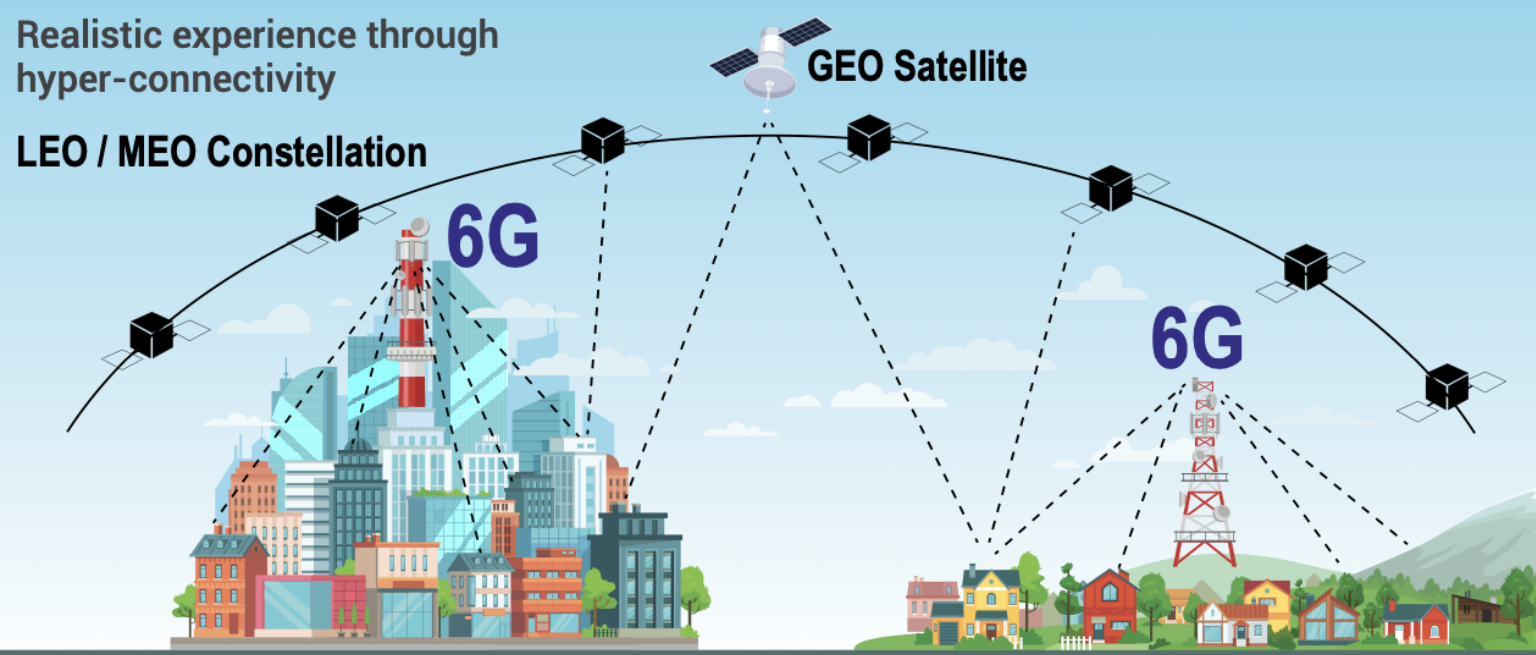
Integrated Sensing and Communication

- Wireless infrastructure performs sensing and communication simultaneously

Wireless Energy Transfer

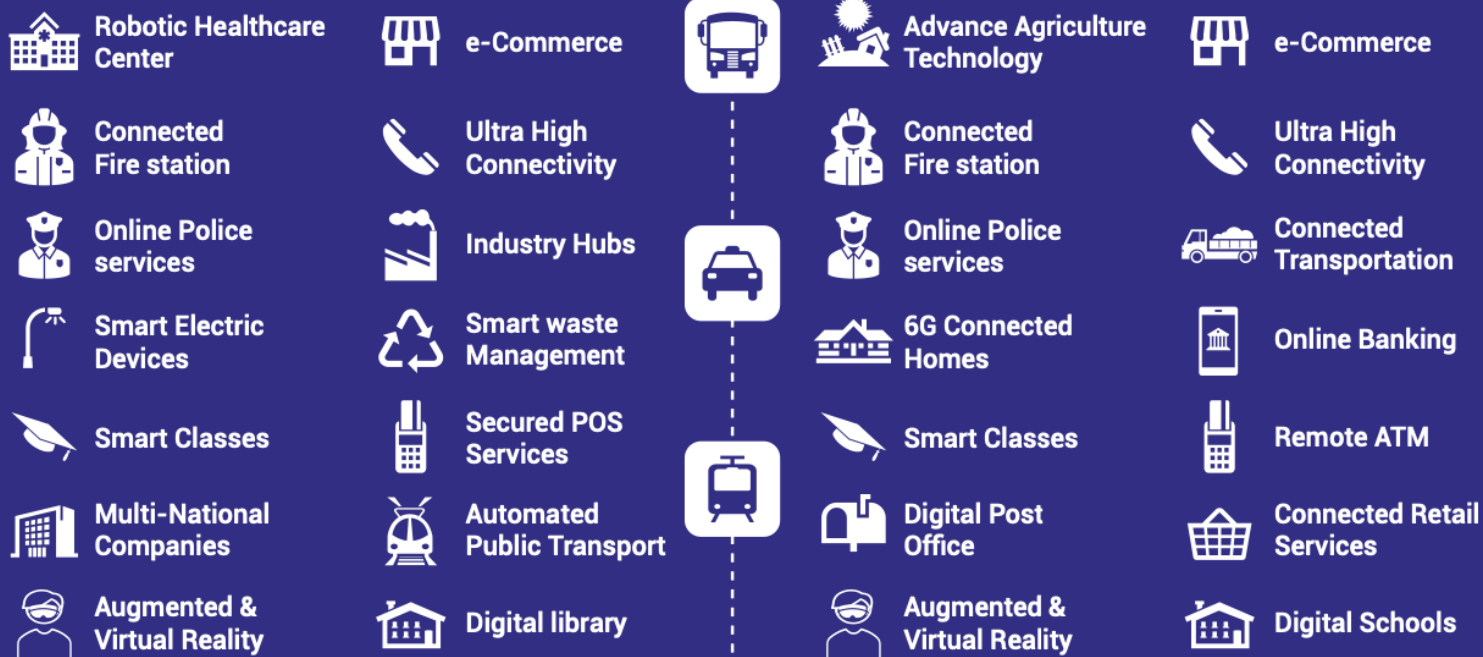
- Simultaneous wireless information and power transmission





Source: [*Bharat 6G Vision Statement*](#),
Department of Telecommunications,
Government of India

UBIQUITOUS CONNECTIVITY



Part II: Open-Source Platforms for IoT Development

RISC-V Microcontroller Ecosystem

- Open instruction set architecture enabling flexible hardware design
- Modular instruction extensions optimized for embedded applications
- Rapid innovation in open hardware ecosystems

Example Platforms

- BL602 and BL604 RISC-V microcontrollers
- BL808 AIoT processors
- GD32V microcontroller family
- CH32V industrial microcontrollers

<https://www.microcontrollertips.com/what-is-risc-v-ecosystem-faq/>

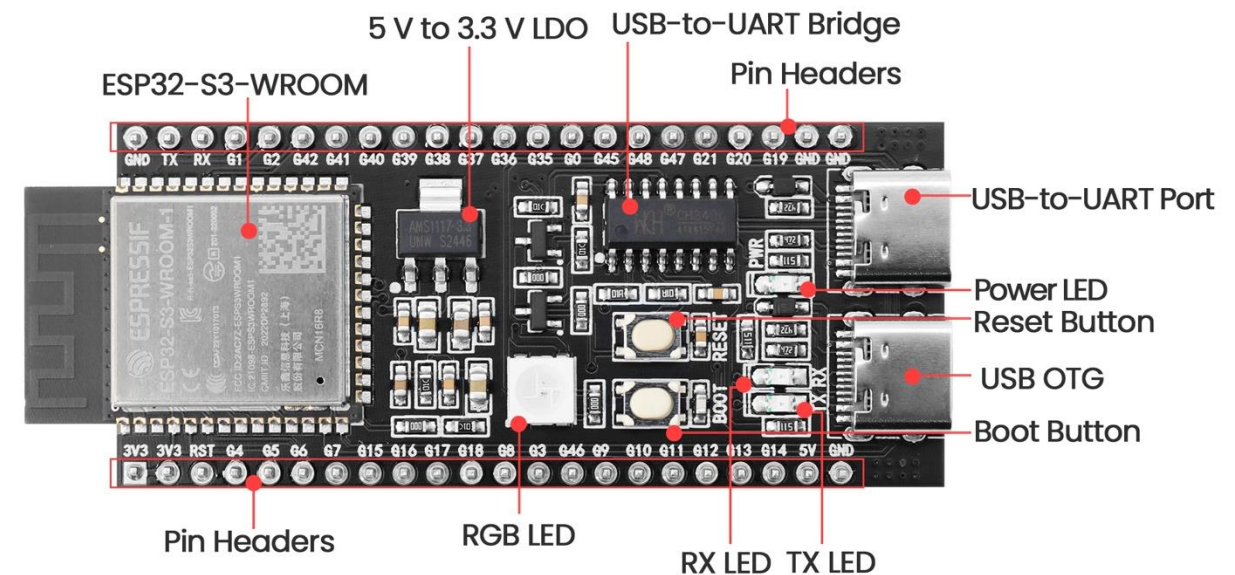
ESP32 Development Ecosystem

Key Variants

- ESP32-S3 optimized for edge AI applications
- ESP32-C3 based on low-power RISC-V architecture
- ESP32-C6 supporting Wi-Fi 6 and Thread protocols

Typical Applications

- Smart home devices
- IoT gateways
- Edge computing prototypes



Edge Computing Boards



Example Platforms

- Raspberry Pi platforms
- Orange Pi systems
- Rock Pi boards
- BeagleBone embedded platforms

Capabilities

- Edge analytics
- Gateway processing
- Robotics and intelligent control



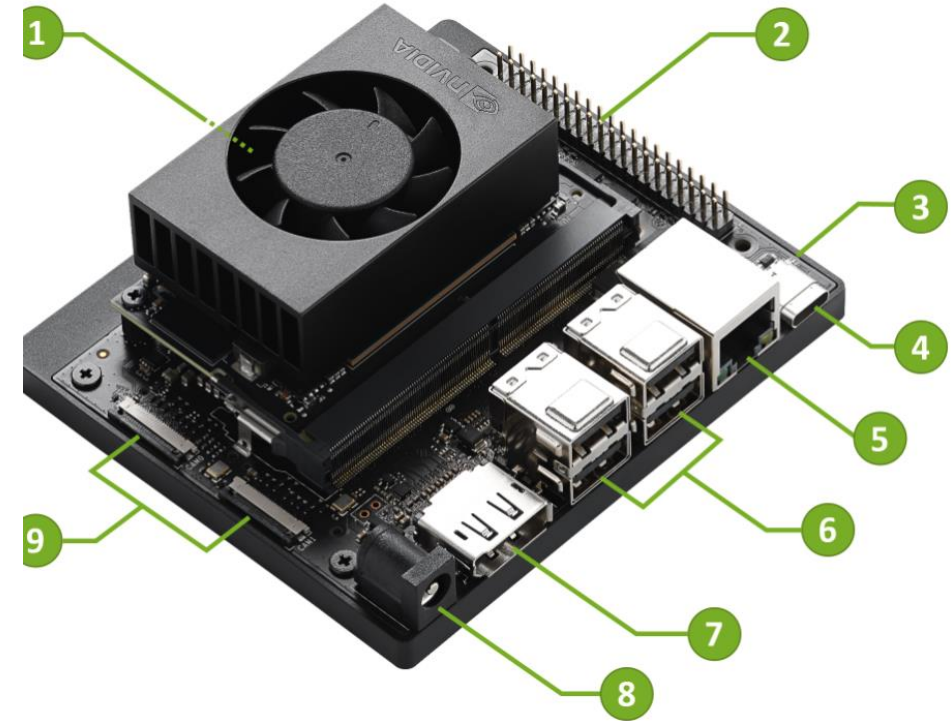
AI-Accelerated Edge Hardware

Example Platforms

- NVIDIA Jetson series
- Google Coral Edge TPU

Application Areas

- Real-time computer vision
- Robotics and autonomous systems
- Edge AI inference



- | | |
|--------------------------------------|--------------------------------------|
| 1 microSD card slot for main storage | 6 USB 3.1 Type A ports (x4) |
| 2 40-pin expansion header | 7 DisplayPort connector |
| 3 Power indicator LED | 8 DC Barrel jack for 19V power input |
| 4 USB-C port for data only | 9 MIPI CSI camera connectors |
| 5 Gigabit Ethernet port | |

FreeRTOS

- Minimal microkernel architecture
- Extremely small memory footprint
- Deterministic real-time scheduling

Zephyr RTOS

- Modular operating system architecture
- Built-in networking stack
- Memory protection and device driver framework

IoT Middleware Platforms



ThingsBoard

- Device telemetry ingestion
- Real-time dashboards
- Rule-based data processing

Mainflux

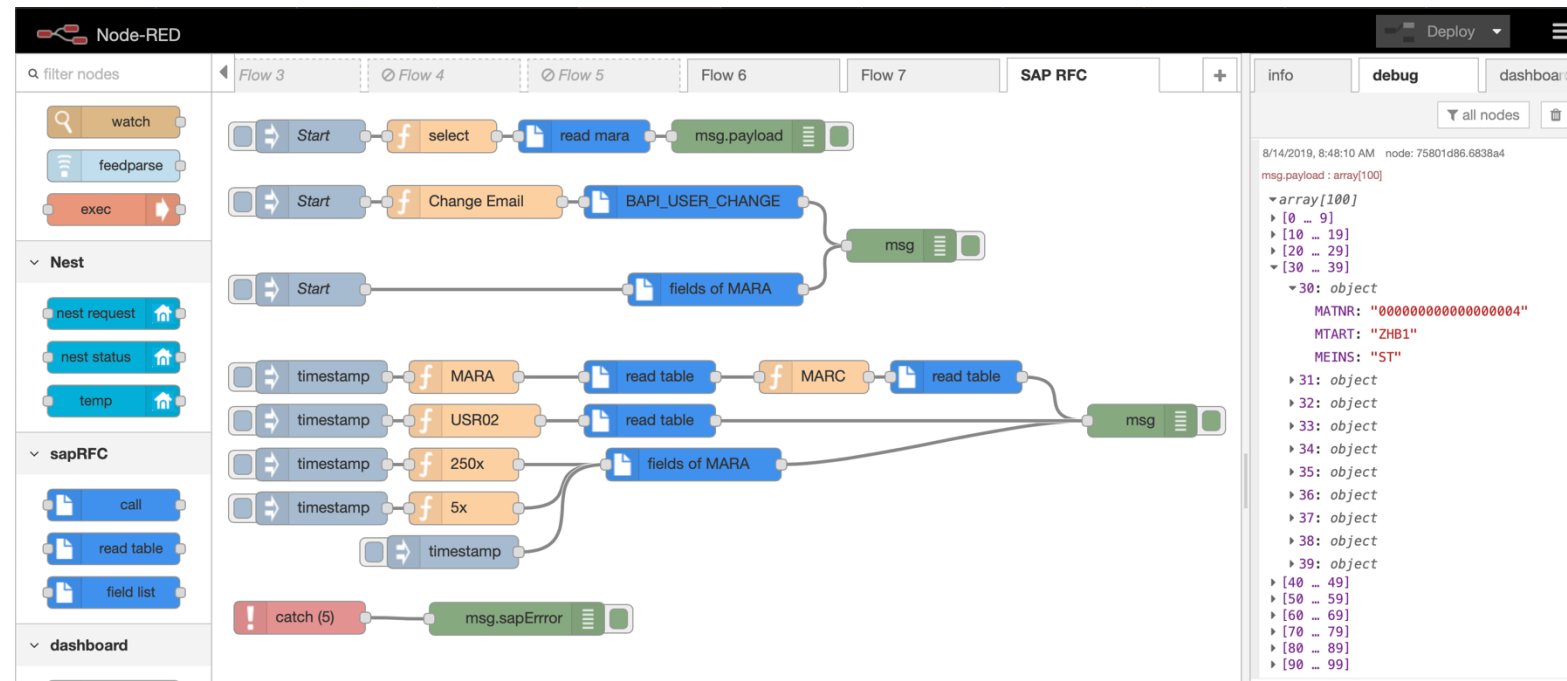
- Microservices architecture
- High-performance messaging
- Secure device authentication

Visual IoT Development Tools

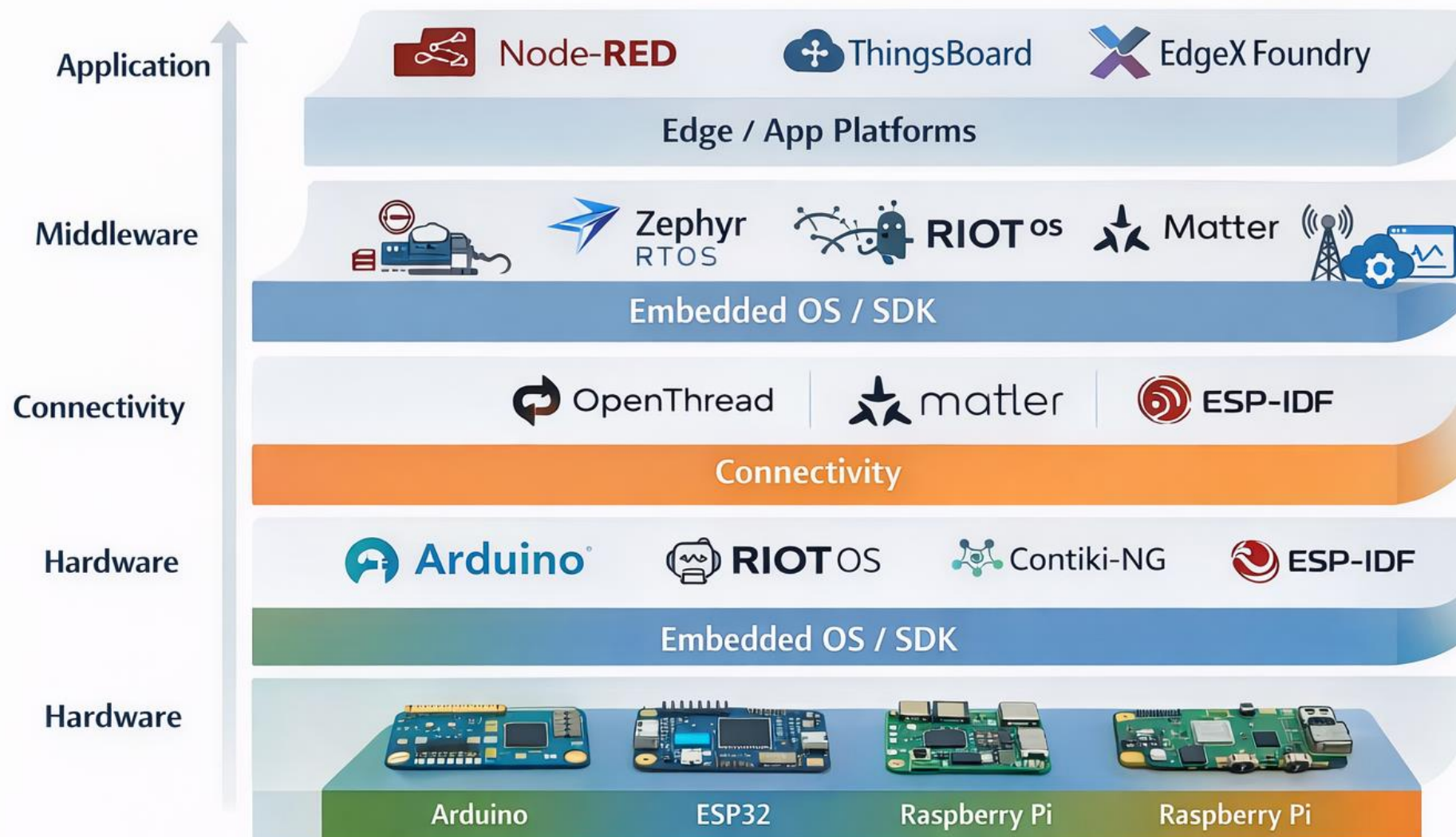


Node-RED

- Flow-based programming for IoT applications
- Rapid prototyping of device interactions
- Integration with cloud services and APIs



IoT Open-Source Development Stack



awesome-open-iot!!



<https://github.com/Agile-IoT/awesome-open-iot>

Gateway

Name	Description	License	Language
Eclipse Kura  Star 557	Eclipse Kura is a Java/OSGi-based framework for IoT gateways		Java
OpenWrt	OpenWrt is an embedded operating system based on the Linux kernel, primarily used on embedded devices to route network traffic	GPLv2	C, Lua
UBOS  Star 16	UBOS is an Arch-derived Linux distro optimized for cost-efficient administration Personal Servers and Indie IoT Devices	Linux	many

Home Automation

Name	Description	License	Language
Calaos  Star 162	Calaos is a free software project (GPLv3) that lets you control and monitor your home.	GPLv3	C++
Freedomotic  Star 429	Freedomotic is an open source, flexible, secure Internet of Things (IoT) development framework, useful to build and manage modern smart spaces.	GPLv2	Java

MQTT and Sparkplug

- Standard messaging transport protocol
- Structured payload definitions for industrial IoT
- Automatic device discovery and state management

Matter Protocol

- Unified interoperability standard for smart home ecosystems
- Communication via Wi-Fi and Thread networks
- Vendor-agnostic device integration

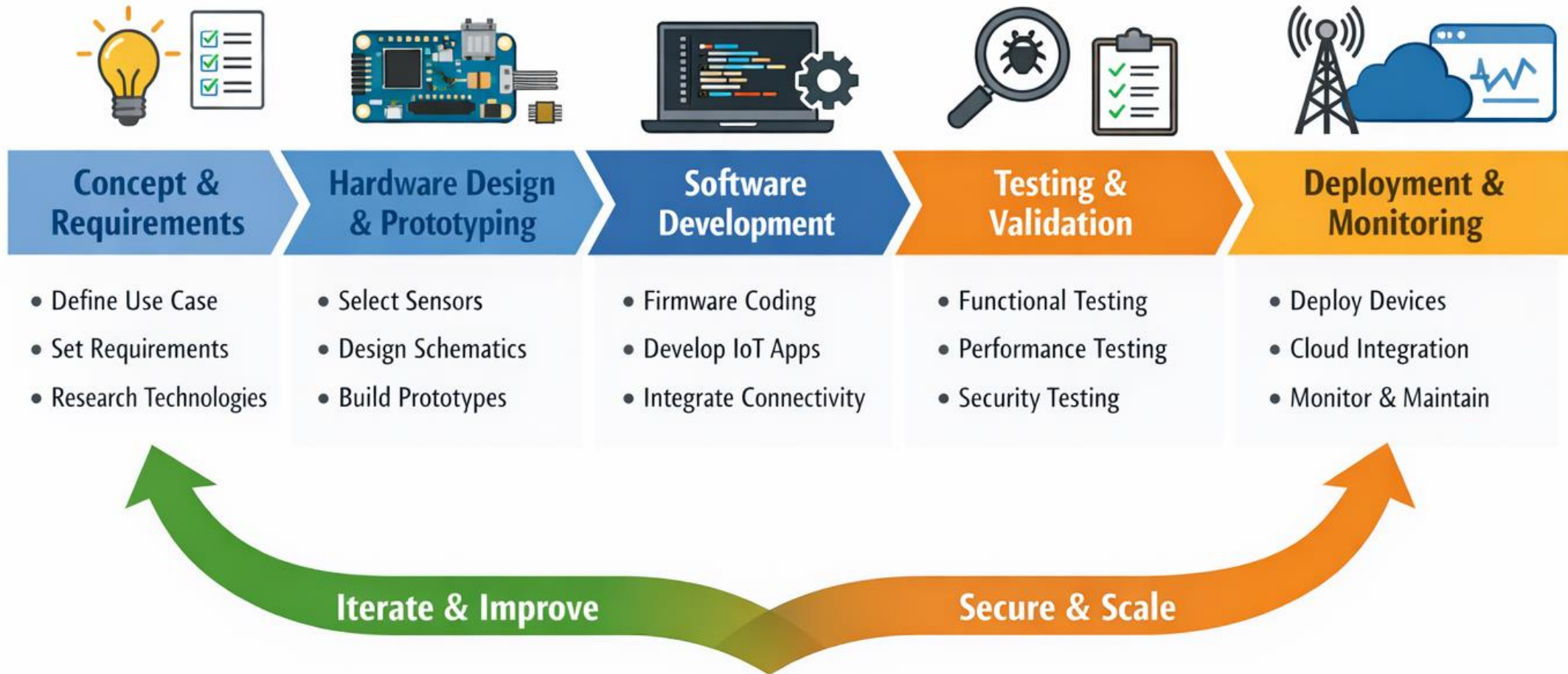
Secure Software Updates

- Firmware updates
- Containerized application updates
- AI model updates

Update Techniques

- A/B partitioning for safe updates
- Cryptographic verification mechanisms
- Rollback protection to prevent device failure

IoT Development Process



Industrial Deployment Challenges



Operational Requirements

- Long device lifecycle management
- Hardware supply stability
- Continuous security monitoring

Key Takeaways



Major Research Trends

- Edge intelligence and AI-native IoT
- Federated and distributed machine learning
- Digital twin systems for complex infrastructures
- Zero-trust security architecture
- Sustainable and green IoT systems
- Advanced wireless technologies for future IoT networks

Open-Source Ecosystem

- Hardware platforms such as ESP32 and RISC-V microcontrollers
- Operating systems including Zephyr and FreeRTOS
- Middleware platforms such as ThingsBoard and Mainflux
- Development tools such as Node-RED
- Interoperability standards including MQTT and Matter

NEXT WEEK: 1st Presentation!

- [1] Y. Pan, L. Lei, G. Shen, X. Zhang and P. Cao, "A Survey on Digital Twin Networks: Architecture, Technologies, Applications, and Open Issues," in *IEEE Internet of Things Journal*, vol. 12, no. 12, pp. 19119-19143, Jun. 2025, doi: 10.1109/JIOT.2025.3565265
- [2] J. Lee, F. Solat, T. Y. Kim and H. V. Poor, "Federated Learning-Empowered Mobile Network Management for 5G and Beyond Networks: From Access to Core," in *IEEE Communications Surveys & Tutorials*, vol. 26, no. 3, pp. 2176-2212, thirdquarter 2024
- [3] D. C. Nguyen *et al.*, "6G Internet of Things: A Comprehensive Survey," in *IEEE Internet of Things Journal*, vol. 9, no. 1, pp. 359-383, 1 Jan.1, 2022, doi: 10.1109/JIOT.2021.3103320